

TITLE

Advancing Harmlessness in AI Assistants Through Constitutional AI and AI Feedback

ABSTRACT

This research explores Constitutional AI (CAI) and Reinforcement Learning from AI Feedback (RLAIF) as novel methods for training AI assistants to be harmless without relying on human labels for harmfulness. We investigate the efficacy of CAI, combining supervised learning (SL) and reinforcement learning (RL) phases, and compare it with Reinforcement Learning from Human Feedback (RLHF). Datasets include human-written prompts for red teaming and helpfulness, alongside model-generated prompts. Key findings demonstrate CAI's ability to train harmless, non-evasive AI. RLAIF models show preference over RLHF models for harmlessness and minimal cost to helpfulness. Chain-of-thought reasoning further enhances harm identification.

1. Introduction

The development of capable AI assistants necessitates robust mechanisms to ensure harmlessness. Traditional approaches, such as Reinforcement Learning from Human Feedback (RLHF), rely heavily on human annotations, which can be costly and may not adequately cover the vast spectrum of potential harms. This paper investigates Constitutional AI (CAI) as a framework to address these limitations. CAI leverages AI feedback to guide the training process, aiming to produce harmless and helpful AI assistants. We present experiments that compare CAI, particularly its Reinforcement Learning from AI Feedback (RLAIF) variant,

with RLHF and supervised learning (SL) based methods. The objective is to assess the effectiveness of CAI in achieving high harmlessness scores while maintaining or even improving helpfulness, and to explore the nuances of AI-driven critique and revision for harm reduction.

2. Methods

2.1 Datasets

A variety of datasets were utilized to train and evaluate the AI assistants. These include: human-written prompts specifically designed for red teaming (42,496 instances) and human-written prompts focused on helpfulness (135,296 instances). Model-generated prompts were created via few-shot prompting a pre-trained model (140,335 instances). Additional data comprised human feedback data for preference model comparisons collected in prior work, and helpfulness and harmlessness data collected separately by crowdworkers. For harm-focused multiple-choice evaluations, we utilized a dataset from [Ganguli et al., 2022]. Specific prompt sets from established models such as PALMS, InstructGPT, and LaMDA were also incorporated. To gauge real-world perception, a survey of 399 undergraduate and postgraduate students from various disciplines in Hong Kong was conducted.

2.2 Algorithms

The core methodologies employed include: Constitutional AI (CAI), which involves a supervised learning phase (SL) and a reinforcement learning phase (RL). Within CAI, we specifically explored Reinforcement Learning from AI Feedback (RLAIF). Other methods investigated for comparison or as components include: Scaling

Supervision, Chain-of-thought (CoT) reasoning, Reinforcement Learning from Human Feedback (RLHF), a Critique and Revision pipeline, Self-improvement through AI feedback, and Preference model training.

2.3 Analytical Approach

The analytical approach involved several key algorithmic processes. Fine-tuning a pretrained language model on revised responses was performed as a Supervised Learning step. Subsequently, a preference model was trained from AI preferences. The trained preference model served as a reward signal for reinforcement learning using the RLAIIF approach. To enhance harmlessness, we implemented algorithms for generating self-critiques and revisions, and explored a direct revision generation method as an alternative to the critique-revision pipeline. Ensembling over constitutional principles was also investigated. For specific tasks, we employed techniques like clamping probabilities for CoT labels. Underlying generative models included Variational Autoencoder (VAE), Generative Adversarial Networks (GANs), and the autoregressive large language model ChatGPT.

3. Results

3.1 Key Findings

Constitutional AI (CAI) demonstrates the capability to train a harmless yet non-evasive AI assistant without relying on human labels for harmfulness. The Reinforcement Learning from AI Feedback (RLAIIF) approach successfully trains models that are preferred by crowdworkers over models trained using human feedback for harmlessness. Incorporating Chain-of-thought (CoT) reasoning

significantly improves the AI's ability to identify harms, leading to evaluations competitive with preference models trained with human feedback. The model-generated critiques and subsequent revisions show a progressive reduction in harmfulness. Crucially, generating critiques leads to better harmlessness outcomes compared to directly generating revisions. Self-supervised preference labels for RL further enhance model behavior, achieving harmlessness performance equal to or exceeding that of human feedback-based methods. While increasing the number of principles in the constitution does not significantly boost harmlessness scores, it can lead to improved response diversity. Critiqued revisions generally exhibit higher harmlessness scores than direct revisions, particularly for smaller models. RLAIF models are demonstrably more harmless than RLHF and SL-CAI models, with negligible impact on helpfulness. RL-CAI models exhibit virtually no evasiveness and frequently provide nuanced, harmless responses to red team prompts.

3.2 Cross-Paper Comparison

While direct cross-paper comparisons are not explicitly detailed in the provided information, the findings of this work can be contextualized against established literature. Previous research on RLHF (e.g., InstructGPT) has shown its effectiveness in aligning model behavior with human preferences. However, the current study suggests that RLAIF can achieve comparable or superior harmlessness with reduced reliance on human effort. The use of AI-generated critiques and revisions builds upon prior work in self-correction and iterative refinement, demonstrating a scalable pathway to harm reduction. The success of CAI in producing non-evasive responses contrasts with some earlier models that might default to overly cautious or unhelpful answers when faced with

potentially harmful queries. The survey of students in Hong Kong provides a novel empirical validation of user perception of these advanced models.

4. Conclusion

This research underscores the significant potential of Constitutional AI (CAI) and Reinforcement Learning from AI Feedback (RLAIF) in developing highly harmless AI assistants. CAI, by enabling self-critique and revision, effectively mitigates harmful outputs without requiring extensive human labeling for harmfulness. The RLAIF approach, in particular, demonstrates that AI feedback can be a powerful substitute for human feedback in achieving superior harmlessness, with minimal trade-offs in helpfulness. The integration of Chain-of-thought reasoning further refines the AI's ability to detect and avoid harms. These findings represent a promising step towards building safer and more reliable AI systems that can be deployed with greater confidence.

5. References

[Ganguli et al., 2022] (Specific citation details not provided in the input)